ONLINE QUIZ SYSTEM
PHASE 1 – VALIDATION SPECIFICATION

Department: CSE
Subject: Software Engineering
5TH SEM, 'H'
TEAM: 13

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Problem Statement Analysis:

1.1 Problem Statement

Traditional quiz systems often rely on manual preparation, distribution, evaluation, and result calculation. These processes can require significant time and may lead to errors in scoring and result preparation. Managing a large number of questions and participants manually can also make the process difficult to organize.
The Online Quiz System aims to provide a structured platform for conducting multiple-choice quizzes. The system allows quiz participants to answer questions and receive their results, while administrators can manage the questions used in the quizzes.

1.2 Existing System:

Aspect	Existing System
Quiz preparation	Questions are prepared manually
Question distribution	Questions may be provided through paper/forms or manually shared
Answer submission	Answers are manually recorded or submitted through basic forms
Evaluation	Scores may require manual calculation
Result generation	Results may take additional time to prepare
Question management	Questions are maintained manually
Error possibility	Manual evaluation can introduce errors

The existing approach can become inefficient when the number of questions or participants increases. Manual evaluation and question management can also make the process more time-consuming.

1.3 Proposed System:

The proposed Online Quiz System is designed to provide a structured platform for conducting multiple-choice quizzes. A Quiz Participant can start a quiz, view questions, select answers, submit the quiz, and view the resulting score. An Administrator can log in and manage quiz questions by adding, editing, deleting, and viewing questions.

The system is intended to reduce manual effort in quiz management and evaluation by automating question presentation, answer collection, score calculation, and result generation.

Quiz Participant

↓

Start Quiz

↓

View Questions

↓

Select Answers

↓

Submit Quiz

↓

Calculate Score

↓

View Result

Administrator

↓

Login

↓

Manage Questions

↓

Add / Edit / Delete / View

1.4 Objectives:

- To provide a structured platform for conducting multiple-choice quizzes.
- To allow participants to answer quiz questions electronically.
- To automatically calculate quiz scores.
- To display quiz results to participants.
- To allow administrators to add, edit, delete, and view questions.
- To reduce manual effort involved in quiz evaluation.

- To improve the organization and management of quiz questions.

1.5 Scope:

- Participant access to quizzes.
  - Display of multiple-choice questions.
  - Selection and submission of answers.
  - Automatic score calculation.
  - Display of results.
  - Administrator login.
  - Adding quiz questions.
  - Editing quiz questions.
  - Deleting quiz questions.
  - Viewing available questions.

2. Feasibility Study:

A feasibility study is conducted to determine whether the proposed Online Quiz System can be developed and operated successfully within the available technical, economic, operational, and time constraints.

2.1 Technical Feasibility:

Factor	Analysis
Programming	C/C++ can be used for implementing the core quiz functionality
Hardware	A standard computer is sufficient for development and execution
Software	Required compiler/IDE and development tools are readily available
Technical Skills	The project can be developed using programming and database/software engineering concepts known to the team

Complexity	The basic quiz functionality is technically achievable within the project scope
Maintenance	The modular structure allows questions and quiz functionality to be modified later

The proposed Online Quiz System is technically feasible because the required development environment, computing resources, and programming skills are available.

2.2 Economic Feasibility:

Resource	**Estimated Cost**
Development software	Minimal / ₹0 using available tools
Compiler/IDE	₹0 if open-source/free tools are used
Hardware	Existing computer/laptop
Database/software tools	Minimal / ₹0 depending on implementation
Maintenance	Low for the basic system

The project is economically feasible because it can be developed using existing computing resources and freely available development tools. The proposed system does not require significant additional infrastructure for the basic implementation.

2.3 Operational Feasibility:

The proposed Online Quiz System is operationally feasible because its intended operations are straightforward for both participants and administrators.

User	**Operations**
Quiz Participant	Start quiz, view questions, select answers, submit quiz, view result.
Administrator	Login, add questions, edit questions, delete questions, view questions

The system requires minimal training because the participant workflow is simple and the administrator functions are limited to quiz and question management.

2.4 Schedule Feasibility:

Phase	**Phase**	**Estimated Duration**

1	Requirements Analysis	1 Week
2	System Design	1 week
3	Implementation	2 Weeks
4	Testing & Validation	1 Week
5	Documentation	1 Week

The proposed project can be completed within the available academic project schedule by dividing the work into requirements analysis, design, implementation, testing, and documentation phases. The four team members can work in parallel on different project components to reduce development time.
Based on the technical, economic, operational, and schedule analysis, the proposed Online Quiz System is feasible for development within the defined project scope and academic timeline.

3. Member 1 Deliverable Summary:

Deliverable	Status
Problem Statement	Completed
Existing System	Completed
Proposed System	Completed
Objectives	Completed
Scope	Completed
Technical Feasibility	Completed
Economic Feasibility	Completed
Operational Feasibility	Completed
Schedule Feasibility	Completed

This section establishes the problem, proposed solution, objectives, scope, and feasibility of the Online Quiz System. The defined scope and objectives will serve as the foundation for preparing the Software Requirements Specification in the subsequent phase.

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SECTION 2 — SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

2.1 Introduction:

The Online Quiz System is a web-based system used to conduct quizzes online. A user can start a quiz, read the questions, select answers and submit the quiz. After submission, the system calculates the score and shows the result.
The system also has an administrator side. The administrator can log in and manage the questions by adding new questions, changing existing questions, deleting questions and viewing the available questions.
The main purpose of the system is to make the quiz process simple and reduce the need for manual evaluation.

2.2 User Requirements:

The following are the basic requirements from the user's point of view.
UR-01: Start Quiz
The system should allow the user to start an available quiz.
UR-02: View Questions
The user should be able to see the questions and the answer options.
UR-03: Select Answer
The user should be able to select one of the given options as the answer.
UR-04: Navigate Questions
The user should be able to move between the questions while taking the quiz.
UR-05: Submit Quiz
The user should be able to submit the quiz after completing it.
UR-06: View Result
After submission, the user should be able to see the score and result.
UR-07: Exit Quiz
The user should be able to leave the quiz when required.
UR-08: Manage Questions
The administrator should be able to add, edit, delete and view quiz questions.

2.3 System Requirements:

2.3.1 Hardware Requirements

Component	Requirement
Processor	Dual-core processor or above
RAM	Minimum 4 GB

Component	Requirement
Storage	At least 10 GB of free space
Network	Internet or local network connection
Device	Laptop, desktop, tablet or smartphone

2.3.2 Software Requirements:

Component	Requirement
Operating System	Windows / Linux / macOS
Frontend	HTML, CSS and JavaScript
Backend	Node.js or suitable server-side technology
Database	MySQL / MongoDB
Browser	Chrome / Firefox / Edge / Safari
IDE	Visual Studio Code or any similar IDE

2.4 Assumptions:

1. The user has access to a device with a supported web browser.
2. A working internet or local network connection is available.
3. The quiz questions and correct answers are entered by the administrator.
4. Only authorized administrators are allowed to change the questions.
5. The database is available when the system is being used.
6. The first version of the system mainly supports multiple-choice questions.
7. Users are expected to enter valid information while using the system.

2.5 Constraints:

1. The system depends on the availability of the server and network.
2. Only an authorized administrator can add, edit or delete questions.
3. The initial version mainly supports multiple-choice quizzes.
4. The performance of the system can depend on the server and network conditions.
5. A supported web browser is required to access the system.
6. Phase 1 is mainly concerned with planning and validation rather than carrying out actual penetration testing.
7. The accuracy of the result depends on the correct answers stored in the database.

SECTION 3 — REQUIREMENT CLASSIFICATION:

3.1 Functional Requirements (FR)

Functional requirements describe the main functions that the Online Quiz System should provide.

FR-01: Start Quiz
The system shall allow the user to start an available quiz.

FR-02: Display Questions
The system shall display multiple-choice questions along with their answer options.

FR-03: Select Answer
The system shall allow the user to select an answer for each question.

FR-04: Navigate Questions
The system shall allow the user to move between the available quiz questions.

FR-05: Submit Quiz
The system shall allow the user to submit the quiz after completing the questions.

FR-06: Calculate Score
The system shall calculate the user's score based on the answers submitted and the correct answers stored in the system.

FR-07: Display Result
The system shall display the user's score and result after the quiz is submitted.

FR-08: Exit Quiz
The system shall allow the user to exit the quiz.

FR-09: Administrator Login
The system shall allow an administrator to log in using valid credentials.

FR-10: Add Question
The system shall allow an authenticated administrator to add a new quiz question.

FR-11: Edit Question
The system shall allow an authenticated administrator to edit an existing question.

FR-12: Delete Question
The system shall allow an authenticated administrator to delete an existing question.

FR-13: View Questions
The system shall allow an authenticated administrator to view the questions stored in the system.

3.2 Validation/Verification Requirements (VFR):

These requirements are used to check whether the system handles inputs and operations correctly. They also cover basic security, data correctness and usability.

VFR-01: Answer Input Validation
The system shall reject an answer that is not one of the options provided for that question.

VFR-02: Unanswered Question Validation
The system shall identify unanswered questions before the quiz is submitted, according to the quiz rules.
VFR-03: Administrator Authentication
The system shall not allow an unauthorized user to access administrator functions.
VFR-04: Login Validation
The system shall reject an administrator login attempt when the entered credentials are incorrect.
VFR-05: Question Input Validation
The system shall not accept incomplete or invalid question information from the administrator.
VFR-06: Score Verification
The system shall calculate the score correctly by comparing the submitted answers with the correct answers.
VFR-07: Data Integrity Verification
The system shall make sure that adding, editing or deleting questions does not cause incorrect or inconsistent quiz data.
VFR-08: Stored Data Validation
The system shall check the question and answer data before storing it in the database.
VFR-09: Session Security Validation
The system shall prevent unauthorized users from accessing an administrator session.
VFR-10: Error Handling Verification
The system shall show a suitable error message when the user enters invalid data or performs an invalid operation.
VFR-11: Usability Validation
The system shall provide clear instructions and an interface that is easy for both users and administrators to understand.
VFR-12: Performance Validation
The system shall provide a reasonable response time when used under the expected normal load.

3.3 FR and VFR Summary:

ID	Requirement	Type
FR-01	Start quiz	FR
FR-02	Display questions	FR
FR-03	Select answer	FR
FR-04	Navigate questions	FR

ID	Requirement	Type
FR-05	Submit quiz	FR
FR-06	Calculate score	FR
FR-07	Display result	FR
FR-08	Exit quiz	FR
FR-09	Administrator login	FR
FR-10	Add question	FR
FR-11	Edit question	FR
FR-12	Delete question	FR
FR-13	View questions	FR
VFR-01	Validate answer input	VFR
VFR-02	Validate unanswered questions	VFR
VFR-03	Check administrator access	VFR
VFR-04	Validate login credentials	VFR
VFR-05	Validate question inputs	VFR
VFR-06	Verify score calculation	VFR
VFR-07	Verify data integrity	VFR
VFR-08	Validate stored data	VFR
VFR-09	Check session security	VFR
VFR-10	Verify error handling	VFR
VFR-11	Validate usability	VFR
VFR-12	Validate performance	VFR

3.4 Difference Between FR and VFR:

Functional Requirements explain **what the system is supposed to do**.
Validation/Verification Requirements explain **how we check that the system works correctly and handles incorrect situations properly**.
For example:
FR-03: The system shall allow the user to select an answer.
VFR-01: The system shall reject an answer that is not one of the options provided.
Therefore, FRs describe the functions of the system, while VFRs are used to check the correctness, security and proper behaviour of those functions.

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Section 4: Requirements Traceability Matrix (RTM):

4.1 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) establishes a relationship between system requirements, test cases, and validation methods. It ensures that every requirement can be verified and that no requirement is missed during testing.

Requirement ID	Requirement	Type	Test Case	Validation Method
FR-01	The system shall allow the user to start a quiz.	FR	TC-01	Functional Testing
FR-02	The system shall display multiple-choice questions.	FR	TC-02	Functional Testing
FR-03	The system shall allow the user to select an answer.	FR	TC-03	Functional Testing
FR-04	The system shall allow the user to submit the quiz.	FR	TC-04	Functional Testing
FR-05	The system shall calculate the user's score.	FR	TC-05	Functional Testing
FR-06	The system shall display the quiz result to the user.	FR	TC-06	Functional Testing
FR-07	The system shall allow the administrator to log in.	FR	TC-07	Functional Testing
FR-08	The system shall allow the administrator to add questions.	FR	TC-08	Functional Testing
FR-09	The system shall allow the administrator to edit questions.	FR	TC-09	Functional Testing
FR-10	The system shall allow the administrator to delete questions.	FR	TC-10	Functional Testing
VFR-01	The system shall reject invalid or empty answer inputs.	VFR	TC-11	Input Validation / Boundary Testing
VFR-02	The system shall prevent unauthorized users from accessing administrator functions.	VFR	TC-12	Security Testing
VFR-03	The system shall prevent submission of incomplete required information.	VFR	TC-13	Input Validation
VFR-04	The system shall reject invalid administrator login credentials.	VFR	TC-14	Security Testing
VFR-05	The system shall display appropriate error messages for invalid inputs.	VFR	TC-15	Usability / Input Validation

4.2 Test Case Summary:

Test Case ID	Test Case Description	Expected Result
TC-01	Click on "Start Quiz".	The quiz should start successfully.
TC-02	Start a quiz and view the questions.	Multiple-choice questions should be displayed.
TC-03	Select an answer for a question.	The selected answer should be recorded.
TC-04	Click on "Submit Quiz".	The quiz should be submitted successfully.
TC-05	Submit a completed quiz.	The system should calculate the correct score.
TC-06	Complete the quiz.	The score/result should be displayed.
TC-07	Enter valid administrator credentials.	The administrator should be logged in.
TC-08	Add a new question.	The question should be added successfully.
TC-09	Edit an existing question.	The question should be updated successfully.
TC-10	Delete an existing question.	The question should be deleted successfully.
TC-11	Submit without providing a required answer.	The system should reject the input and display a validation message.
TC-12	Attempt to access administrator functions without authorization.	Access should be denied.
TC-13	Submit incomplete required information.	The system should reject the submission.
TC-14	Enter incorrect administrator login credentials.	Login should fail and an appropriate message should be displayed.
TC-15	Enter invalid input in a form field.	The system should reject or appropriately handle the input.

Section 5: Validation Plan:

The validation plan defines the testing activities that will be carried out to ensure that the Online Quiz System satisfies its specified requirements and behaves correctly under valid and invalid conditions. Since Phase 1 focuses on planning, the testing activities described below are planned for subsequent phases.

5.1 Functional Testing:

Functional testing will verify that all major functions of the Online Quiz System operate according to the specified requirements.
The following functions will be tested:

- Starting a quiz
- Displaying multiple-choice questions
- Selecting answers
- Submitting a quiz
- Calculating the score
- Displaying quiz results

- Administrator login
- Adding questions
- Editing questions
- Deleting questions

5.2 Non-Functional Testing:

Non-functional testing will evaluate the quality attributes of the system.
The following aspects will be considered:

- **Performance:** Verify that quiz pages and questions load within an acceptable time.
- **Usability:** Verify that users can easily navigate through the quiz and understand the interface.
- **Reliability:** Verify that the system performs consistently without unexpected failures.
- **Compatibility:** Verify that the system functions correctly on supported browsers and devices.

5.3 Security Testing:

Security testing will verify that the system protects restricted functionality and prevents unauthorized access.
The following areas will be tested:

- Administrator authentication
- Unauthorized access to administrator functions
- Invalid administrator login attempts
- User and administrator access control
- Session management
- Protection of quiz and administrator data

For example, if a normal quiz participant attempts to access an administrator page without authorization, the system should deny access.

5.4 Input Validation Testing

Input validation testing will verify that the system correctly handles invalid, empty, or incomplete inputs.
The following cases will be tested:

- Empty answer submission
- Empty question fields
- Missing answer options
- Invalid login credentials
- Invalid or unexpected input values
- Incomplete form submissions

The system should reject invalid inputs and display appropriate validation messages

5.5 Usability Testing:

Usability testing will verify that the Online Quiz System is simple and easy to use.
The following aspects will be evaluated:

- Clear navigation
- Clearly labelled buttons

- Readable questions and answer options
- Simple quiz submission process
- Easy access to quiz results
- Clear and understandable error messages

A first-time user should be able to start, complete, and submit a quiz without requiring technical assistance.

5.6 Basic Penetration-Testing Plan:

Basic penetration testing will be planned to identify common security vulnerabilities in the Online Quiz System. The testing will focus on authentication, authorization, input handling, and session management.

The following security checks are planned:

1. **Unauthorized Access:** Attempt to access administrator pages without logging in. Access should be denied.
2. **Invalid Login Attempts:** Attempt to log in using incorrect credentials. The system should reject the login attempt.
3. **Input Manipulation:** Enter unexpected or malicious input into form fields. The system should safely handle or reject such input.
4. **Session Security:** Attempt to access protected pages after logging out. Access should be denied.
5. **Access Control:** Verify that normal users cannot perform administrator operations.

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SECTION 5 — ACTORS

5.1 Actors in the Online Quiz System:

An actor is a person or external entity that interacts with the system to perform a specific function. The Online Quiz System has two main actors: the Quiz Participant/User and the Administrator.

5.1.1 Quiz Participant/User

The Quiz Participant is the user who takes the online quiz. The participant interacts with the system to answer multiple-choice questions, navigate between questions, submit the quiz, and view the final score and result.

The main use cases of the Quiz Participant are:

- Start Quiz
- View Question
- Select Answer
- Navigate Questions
- Submit Quiz
- View Score
- View Result
- Exit Quiz

5.1.2 Administrator:

The Administrator is responsible for managing the quiz questions stored in the system. The administrator must log in using valid credentials before accessing administrative functions.

The main use cases of the Administrator are

- Login
- View Questions
- Add Question
- Edit Question
- Delete Question

5.2 Actor Summary:

Actor	Description
Quiz Participant/User	Starts a quiz, views questions, selects answers, navigates between questions, submits the quiz, and views the score and result.
Administrator	Logs in and manages the quiz questions by adding, editing, deleting, and viewing questions.

SECTION 6 — USE CASES

6.1 Quiz Participant Use Cases

The following use cases describe the functions available to the Quiz Participant.

Use Case	Description	Related Requirement
Start Quiz	Allows the participant to start an available quiz.	FR-01
View Question	Displays multiple-choice questions along with their answer options.	FR-02
Select Answer	Allows the participant to select an answer from the available options.	FR-03
Navigate Questions	Allows the participant to move between the available quiz questions.	FR-04
Submit Quiz	Allows the participant to submit the completed quiz.	FR-05
View Score	Displays the score calculated from the submitted answers.	FR-06
View Result	Displays the final quiz result after submission.	FR-07
Exit Quiz	Allows the participant to leave the quiz.	FR-08

6.1.1 Participant Use Case Flow

The normal flow of a quiz is:
Start Quiz → View Question → Select Answer → Navigate Questions → Submit Quiz → Calculate Score → View Score → View Result → Exit Quiz
During submission, the system checks for unanswered questions according to the quiz rules. If unanswered questions are found, the system displays a warning before the quiz is submitted, as applicable.

6.2 Administrator Use Cases

The following use cases describe the functions available to the Administrator.

Use Case	Description	Related Requirement
Login	Allows the administrator to access administrative functions using valid credentials.	FR-09
View Questions	Allows the administrator to view the questions stored in the system.	FR-13
Use Case	**Description**	**Related Requirement**
Add Question	Allows the administrator to add a new multiple-choice question.	FR-10
Edit Question	Allows the administrator to modify an existing question.	FR-11
Delete Question	Allows the administrator to delete an existing question.	FR-12

6.2.1 Administrator Use Case Flow

The normal administrator flow is:
Login → Validate Login Credentials → Verify Administrator Access → View Questions → Add / Edit / Delete Question → Exit Administrative Session
The administrator can add, edit, or delete questions only after successful authentication. The system validates question information before storing changes and maintains data consistency.

6.3 Use Case Relationships

The UML Use Case Diagram uses <<include>> and <<extend>> relationships to represent common functions and optional or conditional behaviour.

6.3.1 Include Relationships:

An <<include>> relationship represents a function that is required as part of another use case.

Main Use Case	Included Use Case	Reason
Select Answer	Validate Answer Input	The selected answer must be one of the options provided for the question.
Submit Quiz	Validate Unanswered Questions	The system checks whether any questions remain unanswered according to the quiz rules.
Submit Quiz	Calculate Score	The system calculates the score using the submitted answers and stored correct answers.
View Result	Display Score	The participant's score is displayed as part of the quiz result.

Login	Validate Login Credentials	The administrator's credentials must be checked before access is granted.
Add Question	Validate Question Input	The new question and its answer options must be checked before storage.
Edit Question	Validate Question Input	The modified question information must be checked before storage.
Add Question	Verify Data Integrity	The system checks that adding a question does not create inconsistent data.
Edit Question	Verify Data Integrity	The system checks that editing a question does not create inconsistent data.
Delete Question	Verify Data Integrity	The system checks that deleting a question does not leave inconsistent data.

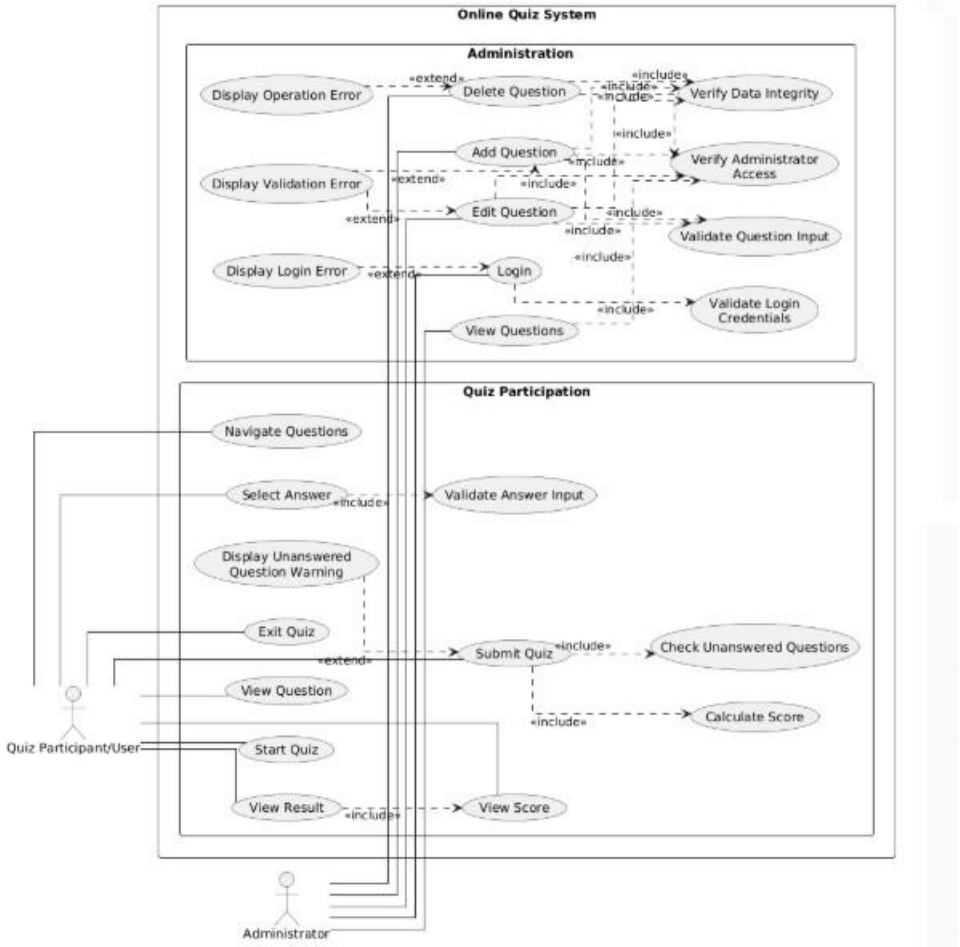
6.3.2 Extend Relationships

An <<extend>> relationship represents additional behaviour that occurs only under a specific condition.

Base Use Case	Extending Use Case	Condition
Submit Quiz	Display Unanswered Question Warning	Extends the submission process when unanswered questions are found.
Login	Display Login Error	Extends the login process when the entered credentials are incorrect.
Add Question	Display Validation Error	Extends the add-question process when the entered information is invalid.
Edit Question	Display Validation Error	Extends the edit-question process when the entered information is invalid.
Delete Question	Display Operation Error	Extends the delete-question process when the operation cannot be completed.

These relationships represent the validation and error-handling behaviour described in the Validation/Verification Requirements. They do not introduce additional major actors or unrelated system features.

SECTION 7 — UML USE CASE DIAGRAM



7.1
Description of the Use Case Diagram

The UML Use Case Diagram represents the interactions between the two main actors and the Online Quiz System.
The Quiz Participant interacts with the system to start a quiz, view multiple-choice questions,

select answers, navigate between questions, submit the quiz, and view the score and final result.

During answer selection and quiz submission, the system performs the required validation and score calculation.

The Administrator interacts with the system by logging in and managing quiz questions. The administrator can view, add, edit, and delete questions after successful authentication. The system validates login credentials and question information, checks data integrity, and displays suitable error messages when invalid information or operations are encountered.

The <<include>> relationships represent mandatory supporting functions, such as answer validation, score calculation, and login validation. The <<extend>> relationships represent conditional behaviour, such as displaying warnings for unanswered questions or showing errors when invalid information is entered.

The diagram is based on the Functional Requirements (FR-01 to FR-13) and the relevant Validation/Verification Requirements (VFR-01 to VFR-10) defined in the Software Requirements Specification.